

# 33

CHAPTER



## Energy and Commodity Derivatives

The variable underlying a derivative is sometimes simply referred to as the *underlying*. Earlier parts of this book have focused on situations where the underlying is a stock price, a stock index, an exchange rate, a bond price, an interest rate, or the loss from a credit event. In this chapter, we consider a variety of other underlyings.

The first part of the chapter is concerned with situations where the underlying is a commodity. Chapter 2 discussed futures contracts on commodities and Chapter 17 discussed how European and American options on commodity futures contracts can be valued. As a European futures option has the same payoff as a European spot option when the futures contract matures at the same time as the option, the model used to value European futures options (Black's model) can also be used to value European spot options. However, American spot options and other more complicated derivatives dependent on the spot price of a commodity require more sophisticated models. A feature of commodity prices is that they often exhibit mean reversion (similarly to interest rates) and are also sometimes subject to jumps. Some of the models developed for interest rates can be adapted to apply to commodities.

The second part of the chapter considers weather and insurance derivatives. A distinctive feature of these derivatives is that they depend on variables with no systematic risk. For example, the expected value of the temperature at a certain location or the losses experienced due to hurricanes can reasonably be assumed to be the same in a risk-neutral world and the real world. This means that historical data is potentially more useful for valuing these types of derivatives than for some others.

### 33.1 AGRICULTURAL COMMODITIES

Agricultural commodities include products that are grown (or created from products that are grown) such as corn, wheat, soybeans, cocoa, coffee, sugar, cotton, and frozen orange juice. They also include products related to livestock such as cattle, hogs, and pork bellies. The prices of agricultural commodities, like all commodities, is determined by supply and demand. The United States Department of Agriculture publishes reports on inventories and production. One statistic that is watched for commodities such as

corn and wheat is the *stocks-to-use ratio*. This is the ratio of the year-end inventory to the year's usage. Typically it is between 20% and 40%. It has an impact on price volatility. As the ratio for a commodity becomes lower, the commodity's price becomes more sensitive to supply changes, so that the volatility increases.

There are reasons for supposing some level of mean reversion in agricultural prices. As prices decline, farmers find it less attractive to produce the commodity and supply decreases creating upward pressure on the price. Similarly, as the price of an agricultural commodity increases, farmers are more likely to devote resources to producing the commodity creating downward pressure on the price.

Prices of agricultural commodities tend to be seasonal, as storage is expensive and there is a limit to the length of time for which a product can be stored. Weather plays a key role in determining the price of many agricultural products. Frosts can decimate the Brazilian coffee crop, a hurricane in Florida is likely to have a big effect on the price of frozen orange juice, and so on. The volatility of the price of a commodity that is grown tends to be highest at pre-harvest times and then declines when the size of the crop is known. During the growing season, the price process for an agricultural commodity is liable to exhibit jumps because of the weather.

Many of the commodities that are grown and traded are used to feed livestock. (For example, the corn futures contract that is traded by the CME Group refers to the corn that is used to feed animals.) The price of livestock, and when slaughtering takes place, is liable to be dependent on the price of these commodities, which are in turn influenced by the weather.

### 33.2 METALS

Another important commodity category is metals. This includes gold, silver, platinum, palladium, copper, tin, lead, zinc, nickel, and aluminum. Metals have quite different characteristics from agricultural commodities. Their prices are unaffected by the weather and are not seasonal. They are extracted from the ground. They are divisible and are relatively easy to store. Some metals, such as copper, are used almost entirely in the manufacture of goods and should be classified as consumption assets. As explained in Section 5.1, others, such as gold and silver, are held purely for investment as well as for consumption and should be classified as investment assets.

As in the case of agricultural commodities, analysts monitor inventory levels to determine short-term price volatility. Exchange rate volatility may also contribute to volatility as the country where the metal is extracted is often different from the country in whose currency the price is quoted. In the long term, the price of a metal is determined by trends in the extent to which a metal is used in different production processes and new sources of the metal that are found. Changes in exploration and extraction methods, geopolitics, cartels, and environmental regulation also have an impact.

One potential source of supply for a metal is recycling. A metal might be used to create a product and, over the following 20 years, 10% of the metal might come back on the market as a result of a recycling process.

Metals that are investment assets are not usually assumed to follow mean-reverting processes because a mean-reverting process would give rise to an arbitrage opportunity for the investor. For metals that are consumption assets, there may be some mean

reversion. As the price of a metal increases, it is likely to become less attractive to use the metal in some production processes and more economically viable to extract the metal from difficult locations. As a result there will be downward pressure on the price. Similarly, as the price decreases, it is likely to become more attractive to use the metal in some production processes and less economically viable to extract the metal from difficult locations. As a result, there will be upward pressure on the price.

### 33.3 ENERGY PRODUCTS

Energy products are among the most important and actively traded commodities. A wide range of energy derivatives trade in both the over-the-counter market and on exchanges. Here we consider oil, natural gas, and electricity. There are reasons for supposing that all three follow mean reverting processes. As the price of a source of energy rises, it is likely to be consumed less and produced more. This creates a downward pressure on prices. As the price of a source of energy declines, it is likely to be consumed more, but production is likely to be less economically viable. This creates upward pressure on the price.

#### Crude Oil

The crude oil market is the largest commodity market in the world, with global demand amounting to about 80 million barrels daily. Ten-year fixed-price supply contracts have been commonplace in the over-the-counter market for many years. These are swaps where oil at a fixed price is exchanged for oil at a floating price.

There are many grades of crude oil, reflecting variations in the gravity and the sulfur content. Two important benchmarks for pricing are Brent crude oil (which is sourced from the North Sea) and West Texas Intermediate (WTI) crude oil. Crude oil is refined into products such as gasoline, heating oil, fuel oil, and kerosene.

In the over-the-counter market, virtually any derivative that is available on common stocks or stock indices is now available with oil as the underlying asset. Swaps, forward contracts, and options are popular. Contracts sometimes require settlement in cash and sometimes require settlement by physical delivery (i.e., by delivery of oil).

Exchange-traded contracts are also popular. The CME Group and Intercontinental-Exchange (ICE) trade a number of oil futures and oil futures options contracts. Some of the futures contracts are settled in cash; others are settled by physical delivery. For example, the Brent crude oil futures traded on ICE have a cash settlement option; the light sweet crude oil futures traded on CME Group require physical delivery. In both cases, the amount of oil underlying one contract is 1,000 barrels. The CME Group also trades popular contracts on two refined products: heating oil and gasoline. In both cases, one contract is for the delivery of 42,000 gallons.

#### Natural Gas

The natural gas industry throughout the world went through a period of deregulation and the elimination of government monopolies in the 1980s and 1990s. The supplier of natural gas is now not necessarily the same company as the producer of the gas. Suppliers are faced with the problem of meeting daily demand.

A typical over-the-counter contract is for the delivery of a specified amount of natural gas at a roughly uniform rate over a 1-month period. Forward contracts, options, and swaps are available in the over-the-counter market. The seller of natural gas is usually responsible for moving the gas through pipelines to the specified location.

The CME Group trades a contract for the delivery of 10,000 million British thermal units of natural gas. The contract, if not closed out, requires physical delivery to be made during the delivery month at a roughly uniform rate to a particular hub in Louisiana. ICE trades a similar contract in London.

Natural gas is a popular source of energy for heating buildings. It is also used to produce electricity, which in turn is used for air-conditioning. As a result, demand for natural gas is seasonal and dependent on the weather.

## Electricity

Electricity is an unusual commodity because it cannot easily be stored.<sup>1</sup> The maximum supply of electricity in a region at any moment is determined by the maximum capacity of all the electricity-producing plants in the region. In the United States there are 140 regions known as *control areas*. Demand and supply are first matched within a control area, and any excess power is sold to other control areas. It is this excess power that constitutes the wholesale market for electricity. The ability of one control area to sell power to another control area depends on the transmission capacity of the lines between the two areas. Transmission from one area to another involves a transmission cost, charged by the owner of the line, and there are generally some transmission or energy losses.

A major use of electricity is for air-conditioning systems. As a result the demand for electricity, and therefore its price, is much greater in the summer months than in the winter months. The nonstorability of electricity causes occasional very large movements in the spot price. Heat waves have been known to increase the spot price by as much as 1,000% for short periods of time.

Like natural gas, electricity has been through a period of deregulation and the elimination of government monopolies. This has been accompanied by the development of an electricity derivatives market. The CME Group now trades a futures contract on the price of electricity, and there is an active over-the-counter market in forward contracts, options, and swaps. A typical contract (exchange-traded or over-the-counter) allows one side to receive a specified number of megawatt hours for a specified price at a specified location during a particular month. In a  $5 \times 8$  contract, power is received for five days a week (Monday to Friday) during the off-peak period (11 p.m. to 7 a.m.) for the specified month. In a  $5 \times 16$  contract, power is received five days a week during the on-peak period (7 a.m. to 11 p.m.) for the specified month. In a  $7 \times 24$  contract, it is received around the clock every day during the month. Option contracts have either daily exercise or monthly exercise. In the case of daily exercise, the option holder can choose on each day of the month (by giving one day's notice) whether to receive the specified amount of power at the specified strike price. When there is monthly exercise a

<sup>1</sup> Electricity producers with spare capacity sometimes use it to pump water to the top of their hydroelectric plants so that it can be used to produce electricity at a later time. This is the closest they can get to storing this commodity.

single decision on whether to receive power for the whole month at the specified strike price is made at the beginning of the month.

An interesting contract in electricity and natural gas markets is what is known as a *swing option* or *take-and-pay option*. In this contract a minimum and maximum for the amount of power that must be purchased at a certain price by the option holder is specified for each day during a month and for the month in total. The option holder can change (or swing) the rate at which the power is purchased during the month, but usually there is a limit on the total number of changes that can be made.

### 33.4 MODELING COMMODITY PRICES

To value derivatives, we are often interested in modeling the spot price of a commodity in the traditional risk-neutral world. From Section 17.7, the expected future price of the commodity in this world is the futures price.

#### A Simple Process

A simple process for a commodity price can be constructed by assuming that the expected growth rate in the commodity price is dependent solely on time and the volatility of the commodity price is constant. The risk-neutral process for the commodity price  $S$  then has the form

$$\frac{dS}{S} = \mu(t) dt + \sigma dz \quad (33.1)$$

and

$$F(t) = \hat{E}[S(t)] = S(0)e^{\int_0^t \mu(\tau) d\tau}$$

where  $F(t)$  is the futures price for a contract with maturity  $t$  and  $\hat{E}$  denotes expected value in a risk-neutral world. It follows that

$$\ln F(t) = \ln S(0) + \int_0^t \mu(\tau) d\tau$$

Differentiating both sides with respect to time gives

$$\mu(t) = \frac{\partial}{\partial t} [\ln F(t)]$$

#### Example 33.1

Suppose that the futures prices of live cattle at the end of July 2008 are (in cents per pound) as follows:

|               |       |
|---------------|-------|
| August 2008   | 62.20 |
| October 2008  | 60.60 |
| December 2008 | 62.70 |
| February 2009 | 63.37 |
| April 2009    | 64.42 |
| June 2009     | 64.40 |

These can be used to estimate the expected growth rate in live cattle prices in a risk-neutral world. For example, when the model in equation (33.1) is used, the

expected growth rate in live cattle prices between October and December 2008, in a risk-neutral world is

$$\ln\left(\frac{62.70}{60.60}\right) = 0.034$$

or 3.4% per 2 months with continuous compounding. On an annualized basis, this is 20.4% per annum.

### Example 33.2

Suppose that the futures prices of live cattle are as in Example 33.1. A certain breeding decision would involve an investment of \$100,000 now and expenditures of \$20,000 in 3 months, 6 months, and 9 months. The result is expected to be that an extra cattle will be available for sale at the end of the year. There are two major uncertainties: the number of pounds of extra cattle that will be available for sale and the price per pound. The expected number of pounds is 300,000. The expected price of cattle in 1 year in a risk-neutral world is, from Example 33.1, 64.40 cents per pound. Assuming that the risk-free rate of interest is 10% per annum, the value of the investment (in thousands of dollars) is

$$-100 - 20e^{-0.1 \times 0.25} - 20e^{-0.1 \times 0.50} - 20e^{-0.1 \times 0.75} + 300 \times 0.644e^{-0.1 \times 1} = 17.729$$

This assumes that any uncertainty about the extra amount of cattle that will be available for sale has zero systematic risk and that there is no correlation between the amount of cattle that will be available for sale and the price.

### Mean Reversion

As already discussed, most commodity prices follow mean-reverting processes. They tend to get pulled back to a central value. A more realistic process than equation (33.1) for the risk-neutral process followed by the commodity price  $S$  is

$$d \ln S = [\theta(t) - a \ln S] dt + \sigma dz \quad (33.2)$$

This incorporates mean reversion and is analogous to the lognormal process assumed for the short-term interest rate in Chapter 30. Note that this process is sometimes written

$$\frac{dS}{S} = [\theta^*(t) - a \ln S] dt + \sigma dz$$

From Itô's lemma, this is equivalent to the process in equation (33.2) when  $\theta^*(t) = \theta(t) + \frac{1}{2}\sigma^2$ .

The trinomial tree methodology in Section 30.7 can be adapted to construct a tree for  $S$  and determine the value of  $\theta(t)$  in equation (33.2) such that  $F(t) = \hat{E}[S(t)]$ . We will illustrate the procedure by building a three-step tree for the situation where the current spot price is \$20 and the 1-year, 2-year, and 3-year futures prices are \$22, \$23, and \$24, respectively. Suppose that  $a = 0.1$  and  $\sigma = 0.2$  in equation (33.2). We first define a variable  $X$  that is initially zero and follows the process

$$dX = -aX dt + \sigma dz \quad (33.3)$$

Using the procedure in Section 30.7, a trinomial tree can be constructed for  $X$ . This is shown in Figure 33.1.

The variable  $\ln S$  follows the same process as  $X$  except for a time-dependent drift. Analogously to Section 30.7, the tree for  $X$  can be converted to a tree for  $\ln S$  by displacing the positions of nodes. This tree is shown in Figure 33.2. The initial node corresponds to a price of 20, so the displacement for that node is  $\ln 20$ . Suppose that the displacement of the nodes at 1 year is  $\alpha_1$ . The values of the  $X$  at the three nodes at the 1-year point are  $+0.3464$ ,  $0$ , and  $-0.3464$ . The corresponding values of  $\ln S$  are  $0.3464 + \alpha_1$ ,  $\alpha_1$ , and  $-0.3464 + \alpha_1$ . The values of  $S$  are therefore  $e^{0.3464+\alpha_1}$ ,  $e^{\alpha_1}$ , and  $e^{-0.3464+\alpha_1}$ , respectively. We require the expected value of  $S$  to equal the futures price. This means that

$$0.1667e^{0.3464+\alpha_1} + 0.6666e^{\alpha_1} + 0.1667e^{-0.3464+\alpha_1} = 22$$

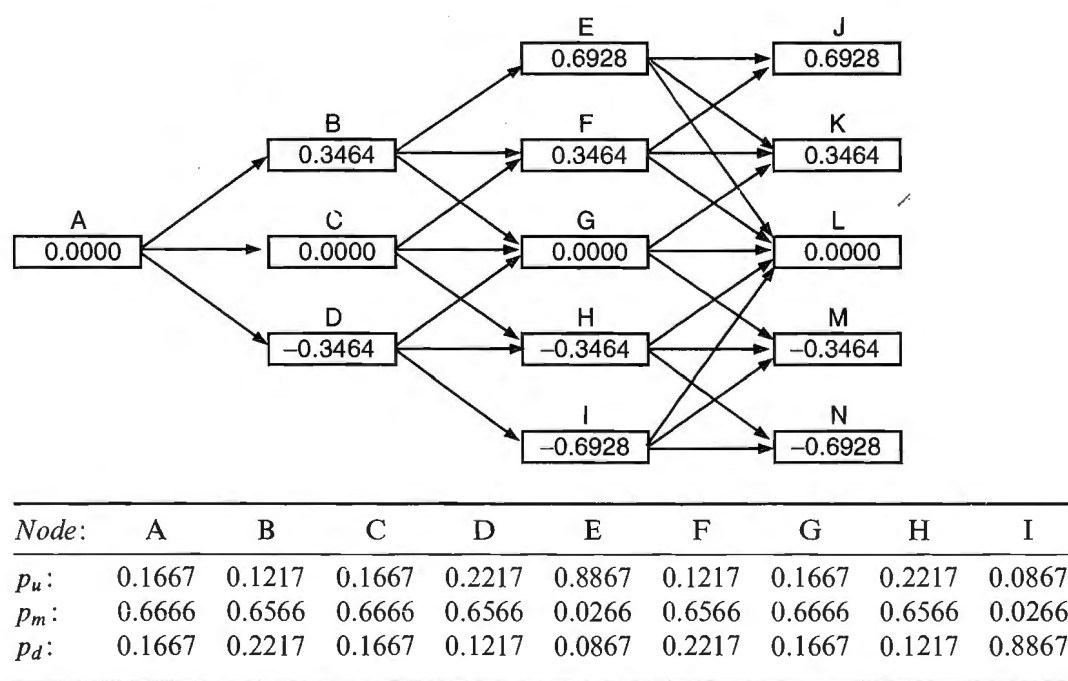
The solution to this is  $\alpha_1 = 3.071$ . The values of  $S$  at the 1-year point are therefore 30.49, 21.56, and 15.25.

At the 2-year point, we first calculate the probabilities of nodes E, F, G, H, and I being reached from the probabilities of nodes B, C, and D being reached. The probability of reaching node F is the probability of reaching node B times the probability of moving from B to F plus the probability of reaching node C times the probability of moving from C to F. This is

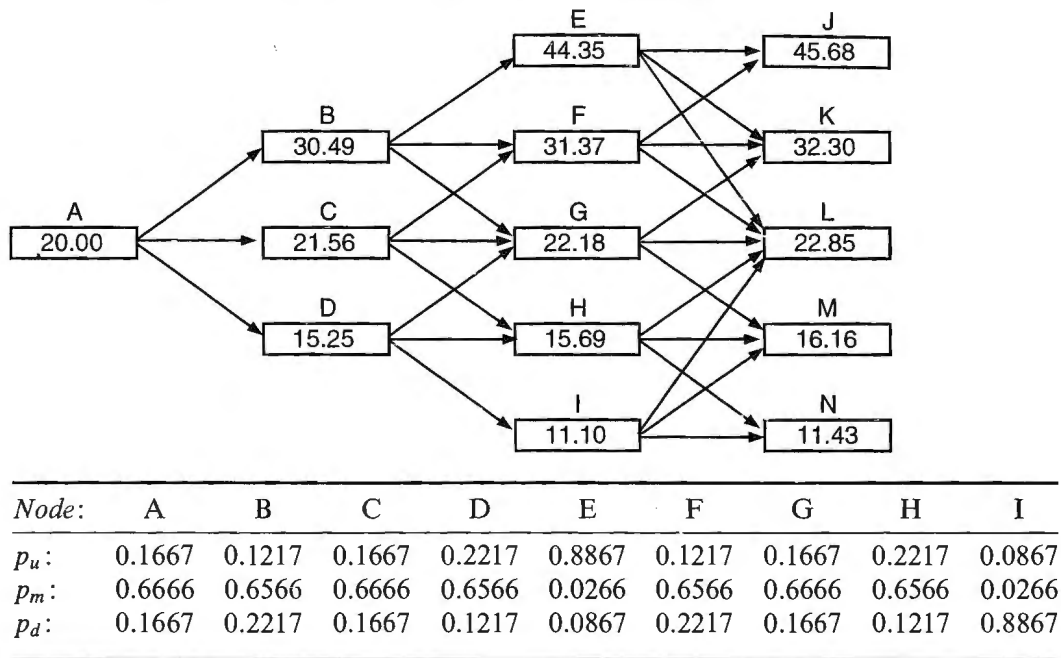
$$0.1667 \times 0.6566 + 0.6666 \times 0.1667 = 0.2206$$

Similarly the probabilities of reaching nodes E, G, H, and I are 0.0203, 0.5183, 0.2206, and 0.0203, respectively. The amount  $\alpha_2$  by which the nodes at time 2 years are

**Figure 33.1** Tree for  $X$ . Constructing this tree is the first stage in constructing a tree for the spot price of a commodity,  $S$ . Here  $p_u$ ,  $p_m$ , and  $p_d$  are the probabilities of “up”, “middle”, and “down” movements from a node.



**Figure 33.2** Tree for spot price of a commodity:  $p_u$ ,  $p_m$ , and  $p_d$  are the probabilities of “up”, “middle”, and “down” movements from a node.



displaced must satisfy

$$0.0203e^{0.6928+\alpha_2} + 0.2206e^{0.3464+\alpha_2} + 0.5183e^{\alpha_2} + 0.2206e^{-0.3464+\alpha_2} + 0.0203e^{-0.6928+\alpha_2} = 23$$

The solution to this is  $\alpha_2 = 3.099$ . This means that the values of  $S$  at the 2-year point are 44.35, 31.37, 22.18, 15.69, and 11.10, respectively.

A similar calculation can be carried out at time 3 years. Figure 33.2 shows the resulting tree for  $S$ .

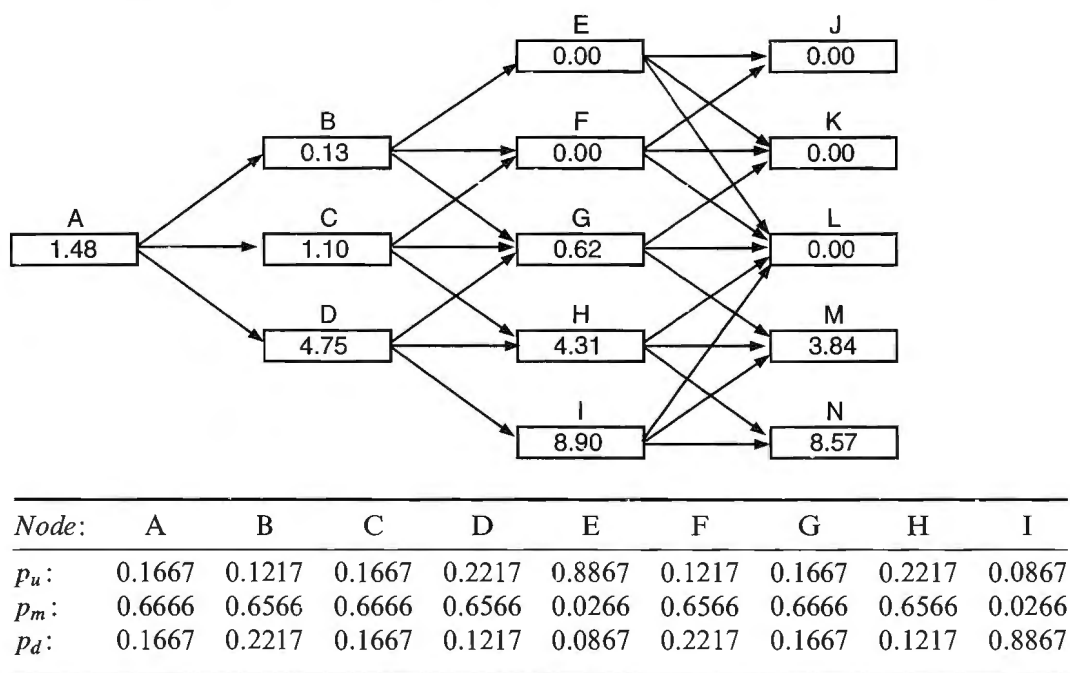
### Example 33.3

Suppose that the tree in Figure 33.2 is used to price a 3-year American put option on the spot price of the commodity with a strike price of 20 when the interest rate (continuously compounded) is 3% per year. Rolling back through the tree in the usual way, we obtain Figure 33.3 showing that the value of the option is \$1.48. The option is exercised early at nodes D, H, and I. To obtain a more accurate value, a tree with many more time steps would be used. The futures prices would be interpolated to obtain futures prices for maturities corresponding to the end of every time step on this more detailed tree.

## Interpolation and Seasonality

When a large number of time steps are used, it is necessary to interpolate between futures prices to obtain a futures price at the end of each time step. When there is seasonality, the interpolation procedure should reflect this. Suppose there are monthly

**Figure 33.3** Valuation of an American put option with a strike price of \$20 using the tree in Figure 33.2.



time steps. One simple way of incorporating seasonality is to collect monthly historical data on the spot price and calculate the 12-month moving average of the price. A *percentage seasonal factor* can then be estimated as the average of the ratio of the spot price for the month to the 12-month moving average of spot prices that is centered (approximately) on the month.

The percentage seasonal factors are then used to deseasonalize the futures prices that are known. Monthly deseasonalized futures are then calculated using interpolation. These futures prices are then seasonalized using the percentage seasonal factors and the tree is built. Suppose, for example, that the futures prices are observed in the market for September and December as 40 and 44, respectively, and we want to calculate a futures prices for October and November. Suppose further that the percentage seasonality factors for September, October, November, and December are calculated from historical data as 0.95, 0.85, 0.8 and 1.1, respectively. The deseasonalized futures prices are  $40/0.95 = 42.1$  for September and  $44/1.1 = 40$  for December. The interpolated deseasonalized futures prices are 41.4 and 40.7 for October and November, respectively. The seasonalized futures prices that would be used in tree construction for October and November are  $41.4 \times 0.85 = 35.2$  and  $40.7 \times 0.8 = 32.6$ , respectively.

As has been mentioned, the volatility of a commodity sometimes shows seasonality. For example, the prices of some agricultural commodities are more volatile during the growing season because of weather uncertainty. Volatility can be monitored using the methods discussed in Chapter 22, and a percentage seasonal factor for volatility can be estimated. The parameter  $\sigma$  can then be replaced by  $\sigma(t)$  in equations (33.2) and (33.3). A procedure that can be used to construct a trinomial tree for the situation

where the volatility is a function of time is discussed in Technical Notes 9 and 16 at [www.rotman.utoronto.ca/~hull/TechnicalNotes](http://www.rotman.utoronto.ca/~hull/TechnicalNotes).

## Jumps

Some commodities, such as electricity and natural gas, exhibit price jumps because of weather-related demand shocks. Other commodities, particularly those that are agricultural, are liable to exhibit price jumps because of weather-related supply shocks. Jumps can be incorporated into equation (33.2) so that the process for the spot price becomes

$$d \ln S = [\theta(t) - a \ln S] dt + \sigma dz + dp$$

where  $dp$  is the Poisson process generating the percentage jumps. This is similar to Merton's mixed jump-diffusion model for stock prices, which is described in Section 26.1. Once the jump frequency and jump size probability distribution have been chosen, the average increase in the commodity price at a future time  $t$  that is as a result of jumps can be calculated. To determine  $\theta(t)$ , the trinomial tree method can be used with the futures prices for maturity  $t$  reduced by this increase. Monte Carlo simulation can be used to implement the model, as explained in Sections 20.6 and 26.1.

## Other Models

More-sophisticated models are sometimes used for oil prices. If  $y$  is the convenience yield, then the proportional drift of the spot price is  $r - y$ , where  $r$  is the short-term risk-free rate and a natural process to assume for the spot price is

$$\frac{dS}{S} = (r - y) dt + \sigma_1 dz_1$$

Gibson and Schwartz suggest that the convenience yield  $y$  be modeled as a mean-reverting process:<sup>2</sup>

$$dy = k(\alpha - y)dt + \sigma_2 dz_2$$

where  $k$  and  $\alpha$  are constants and  $dz_2$  is a Wiener process, which is correlated with the Wiener process  $dz_1$ . To provide an exact fit to futures prices,  $\alpha$  can be made a function of time.

Eydeland and Geman propose a stochastic volatility for gas and electricity prices.<sup>3</sup> This is

$$\begin{aligned} \frac{dS}{S} &= a(b - \ln S) dt + \sqrt{V} dz_1 \\ dV &= c(d - V)dt + e\sqrt{V} dz_2 \end{aligned}$$

where  $a$ ,  $b$ ,  $c$ ,  $d$ , and  $e$  are constants, and  $dz_1$  and  $dz_2$  are correlated Wiener processes. Later Geman proposed a model for oil where the reversion level  $b$  is also stochastic.<sup>4</sup>

<sup>2</sup> See R. Gibson and E. S. Schwartz, "Stochastic Convenience Yield and the Pricing of Oil Contingent Claims," *Journal of Finance*, 45 (1990): 959-76.

<sup>3</sup> A. Eydeland and H. Geman, "Pricing Power Derivatives," *Risk*, September 1998.

<sup>4</sup> H. Geman, "Scarcity and Price Volatility in Oil Markets," EDF Trading Technical Report, 2000.

### 33.5 WEATHER DERIVATIVES

Many companies are in the position where their performance is liable to be adversely affected by the weather.<sup>5</sup> It makes sense for these companies to consider hedging their weather risk in much the same way as they hedge foreign exchange or interest rate risks.

The first over-the-counter weather derivatives were introduced in 1997. To understand how they work, we explain two variables:

HDD: Heating degree days

CDD: Cooling degree days

A day's HDD is defined as

$$\text{HDD} = \max(0, 65 - A)$$

and a day's CDD is defined as

$$\text{CDD} = \max(0, A - 65)$$

where  $A$  is the average of the highest and lowest temperature during the day at a specified weather station, measured in degrees Fahrenheit. For example, if the maximum temperature during a day (midnight to midnight) is 68° Fahrenheit and the minimum temperature is 44° Fahrenheit,  $A = 56$ . The daily HDD is then 9 and the daily CDD is 0.

A typical over-the-counter product is a forward or option contract providing a payoff dependent on the cumulative HDD or CDD during a month. For example, a derivatives dealer could in January 2011 sell a client a call option on the cumulative HDD during February 2012 at the Chicago O'Hare Airport weather station with a strike price of 700 and a payment rate of \$10,000 per degree day. If the actual cumulative HDD is 820, the payoff is \$1.2 million. Often contracts include a payment cap. If the payment cap in our example is \$1.5 million, the contract is the equivalent of a bull spread (see Chapter 11). The client has a long call option on cumulative HDD with a strike price of 700 and a short call option with a strike price of 850.

A day's HDD is a measure of the volume of energy required for heating during the day. A day's CDD is a measure of the volume of energy required for cooling during the day. Most weather derivative contracts are entered into by energy producers and consumers. But retailers, supermarket chains, food and drink manufacturers, health service companies, agriculture companies, and companies in the leisure industry are also potential users of weather derivatives. The Weather Risk Management Association ([www.wrma.org](http://www.wrma.org)) has been formed to serve the interests of the weather risk management industry.

In September 1999 the Chicago Mercantile Exchange (CME) began trading weather futures and European options on weather futures. The contracts are on the cumulative HDD and CDD for a month observed at a weather station. The contracts are settled in cash just after the end of the month once the HDD and CDD are known. One futures contract is on \$20 times the cumulative HDD or CDD for the month. The CME now offers weather futures and options on 42 cities throughout the world. It also offers futures and options on hurricanes, frost, and snowfall.

<sup>5</sup> The US Department of Energy has estimated that one-seventh of the US economy is subject to weather risk.

### 33.6 INSURANCE DERIVATIVES

When derivative contracts are used for hedging purposes, they have many of the same characteristics as insurance contracts. Both types of contracts are designed to provide protection against adverse events. It is not surprising that many insurance companies have subsidiaries that trade derivatives and that many of the activities of insurance companies are becoming very similar to those of investment banks.

Traditionally the insurance industry has hedged its exposure to catastrophic (CAT) risks such as hurricanes and earthquakes using a practice known as reinsurance. Reinsurance contracts can take a number of forms. Suppose that an insurance company has an exposure of \$100 million to earthquakes in California and wants to limit this to \$30 million. One alternative is to enter into annual reinsurance contracts that cover on a pro rata basis 70% of its exposure. If California earthquake claims in a particular year total \$50 million, the costs to the company would then be only \$15 million. Another more popular alternative, involving lower reinsurance premiums, is to buy a series of reinsurance contracts covering what are known as *excess cost layers*. The first layer might provide indemnification for losses between \$30 million and \$40 million; the next layer might cover losses between \$40 million and \$50 million; and so on. Each reinsurance contract is known as an *excess-of-loss* reinsurance contract. The reinsurer has written a bull spread on the total losses. It is long a call option with a strike price equal to the lower end of the layer and short a call option with a strike price equal to the upper end of the layer.<sup>6</sup>

The principal providers of CAT reinsurance have traditionally been reinsurance companies and Lloyds syndicates (which are unlimited liability syndicates of wealthy individuals). In recent years the industry has come to the conclusion that its reinsurance needs have outstripped what can be provided from these traditional sources. It has searched for new ways in which capital markets can provide reinsurance. One of the events that caused the industry to rethink its practices was Hurricane Andrew in 1992, which caused about \$15 billion of insurance costs in Florida. This exceeded the total of relevant insurance premiums received in Florida during the previous seven years. If Hurricane Andrew had hit Miami, it is estimated that insured losses would have exceeded \$40 billion. Hurricane Andrew and other catastrophes have led to increases in insurance/reinsurance premiums.

The over-the-counter market has come up with a number of products that are alternatives to traditional reinsurance. The most popular is a CAT bond. This is a bond issued by a subsidiary of an insurance company that pays a higher-than-normal interest rate. In exchange for the extra interest the holder of the bond agrees to provide an excess-of-loss reinsurance contract. Depending on the terms of the CAT bond, the interest or principal (or both) can be used to meet claims. In the example considered above where an insurance company wants protection for California earthquake losses between \$30 million and \$40 million, the insurance company could issue CAT bonds with a total principal of \$10 million. In the event that the insurance company's California earthquake losses exceeded \$30 million, bondholders would lose some or all of their principal. As an alternative the insurance company could cover this excess cost layer by making a much bigger bond issue where only the bondholders' interest is at risk.

<sup>6</sup> Reinsurance is also sometimes offered in the form of a lump sum if a certain loss level is reached. The reinsurer is then writing a cash-or-nothing binary call option on the losses.

### 33.7 PRICING WEATHER AND INSURANCE DERIVATIVES

One distinctive feature of weather and insurance derivatives is that there is no systematic risk (i.e., risk that is priced by the market) in their payoffs. This means that estimates made from historical data (real-world estimates) can also be assumed to apply to the risk-neutral world. Weather and insurance derivatives can therefore be priced by

1. Using historical data to estimate the expected payoff
2. Discounting the estimated expected payoff at the risk-free rate.

Another key feature of weather and insurance derivatives is the way uncertainty about the underlying variables grows with time. For a stock price, uncertainty grows roughly as the square root of time. Our uncertainty about a stock price in 4 years (as measured by the standard deviation of the logarithm of the price) is approximately twice that in 1 year. For a commodity price, mean reversion kicks in, but our uncertainty about a commodity's price in 4 years is still considerably greater than our uncertainty in 1 year. For weather, the growth of uncertainty with time is much less marked. Our uncertainty about the February HDD at a certain location in 4 years is usually only a little greater than our uncertainty about the February HDD at the same location in 1 year. Similarly, our uncertainty about earthquake losses for a period starting in 4 years is usually only a little greater than our uncertainty about earthquake losses for a similar period starting in 1 year.

Consider the valuation of an option on the cumulative HDD. We could collect 50 years of historical data and estimate a probability distribution for the HDD. This could be fitted to a lognormal or other probability distribution and the expected payoff on the option calculated. This would then be discounted at the risk-free rate to give the value of the option. The analysis could be refined by analyzing trends in the historical data and incorporating weather forecasts produced by meteorologists.

#### Example 33.4

Consider a call option on the cumulative HDD in February 2013 at the Chicago O'Hare Airport weather station with a strike price of 700 and a payment rate of \$10,000 per degree day. Suppose that the HDD is estimated from historical data to have a lognormal distribution with the mean HDD equal to 710 and the standard deviation of the natural logarithm of HDD equal to 0.07. From equation (14A.1), the expected payoff is

$$10,000 \times [710N(d_1) - 700N(d_2)]$$

where

$$d_1 = \frac{\ln(710/700) + 0.07^2/2}{0.07} = 0.2376$$

$$d_2 = \frac{\ln(710/700) - 0.07^2/2}{0.07} = 0.1676$$

or \$250,900. If the risk-free interest rate is 3% and the option is being valued in February 2012 (one year from maturity) the value of the option is

$$250,900 \times e^{-0.03 \times 1} = 243,400$$

or \$243,400.

We might want to adjust the the probability distribution of HDD for temperature trends. Suppose that a linear regression shows that the cumulative HDD for February is decreasing at the rate of 0.5 per year (perhaps because of global warming), so that the estimate of the mean HDD in February 2013 is only 697. Keeping the estimate of the standard deviation of the natural logarithm of the payoff the same, this would reduce the value of the expected payoff to \$180,400 and the value of the option to \$175,100.

Finally, suppose that long-range weather forecasters consider it likely that February 2013 will be particularly mild. The estimate of the expected HDD might then be reduced even further making the option even less valuable.

In the insurance area, Litzenberger *et al.* have shown that there are (as one would expect) no statistically significant correlation between the returns from CAT bonds and stock market returns.<sup>7</sup> This confirms that there is no systematic risk and that valuations can be based on the actuarial data collected by insurance companies.

CAT bonds typically give a high probability of an above-normal rate of interest and a low probability of a big loss. Why would investors be interested in such instruments? The answer is that the expected return (taking account of possible losses) is higher than the return that can be earned on risk-free investments. However, the risk in CAT bonds can (at least in theory) be completely diversified away in a large portfolio. CAT bonds therefore have the potential to improve risk–return trade-offs.

### 33.8 HOW AN ENERGY PRODUCER CAN HEDGE RISKS

There are two components to the risks facing an energy producer. One is the risk associated with the market price for the energy (the price risk); the other is risk associated with the amount of energy that will be bought (the volume risk). Although prices do adjust to reflect volumes, there is a less-than-perfect relationship between the two, and energy producers have to take both into account when developing a hedging strategy. The price risk can be hedged using the energy derivative contracts. The volume risks can be hedged using the weather derivatives. Define:

$Y$ : Profit for a month

$P$ : Average energy prices for the month

$T$ : Relevant temperature variable (HDD or CDD) for the month.

An energy producer can use historical data to obtain a best-fit linear regression relationship of the form

$$Y = a + bP + cT + \epsilon$$

where  $\epsilon$  is the error term. The energy producer can then hedge risks for the month by taking a position of  $-b$  in energy forwards or futures and a position of  $-c$  in weather forwards or futures. The relationship can also be used to analyze the effectiveness of alternative option strategies.

<sup>7</sup> R. H. Litzenberger, D. R. Beaglehole, and C. E. Reynolds, "Assessing Catastrophe Reinsurance-Linked Securities as a New Asset Class," *Journal of Portfolio Management*, Winter 1996: 76–86.

## SUMMARY

When there are risks to be managed, derivatives markets have been very innovative in developing products to meet the needs of the market.

There are a number of different types of commodity derivatives. The underlyings include agricultural products that are grown, livestock, metals, and energy products. The models used to value them usually incorporate mean reversion. Sometimes seasonality is modeled explicitly and jumps are incorporated. Energy derivatives with oil, natural gas, and electricity as the underlying are particularly important and have been the subject of models that are as sophisticated as the most sophisticated models used for stock prices, exchange rates, and interest rates.

In the weather derivatives market, two measures, HDD and CDD, have been developed to describe temperature during a month. These are used to define payoffs on both exchange-traded and over-the-counter derivatives. No doubt, as the weather derivatives market develops, contracts on rainfall, snow, and other weather-related variables will become more common.

Insurance derivatives are an alternative to traditional reinsurance as a way for insurance companies to manage the risk of a catastrophic event such as a hurricane or an earthquake. We may see other sorts of insurance, such as life and automobile insurance, being traded in a similar way in the future.

Weather and insurance derivatives have the property that the underlying variables have no systematic risk. This means that the derivatives can be valued by estimating expected payoffs using historical data and discounting the expected payoff at the risk-free rate.

## FURTHER READING

### ***On commodity derivatives***

Clewlow, L., and C. Strickland. *Energy Derivatives: Pricing and Risk Management*. Lacima Group, 2000.

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Geman, H. "CAT Calls," *Risk* (September 1994): 86–89.

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**Practice Questions (Answers in Solutions Manual)**

- 33.1. What is meant by HDD and CDD?
- 33.2. How is a typical natural gas forward contract structured?
- 33.3. Distinguish between the historical data and the risk-neutral approach to valuing a derivative. Under what circumstance do they give the same answer.
- 33.4. Suppose that each day during July the minimum temperature is 68° Fahrenheit and the maximum temperature is 82° Fahrenheit. What is the payoff from a call option on the cumulative CDD during July with a strike of 250 and a payment rate of \$5,000 per degree-day?
- 33.5. Why is the price of electricity more volatile than that of other energy sources?
- 33.6. Why is the historical data approach appropriate for pricing a weather derivatives contract and a CAT bond?
- 33.7. "HDD and CDD can be regarded as payoffs from options on temperature." Explain this statement.
- 33.8. Suppose that you have 50 years of temperature data at your disposal. Explain carefully the analyses you would carry out to value a forward contract on the cumulative CDD for a particular month.
- 33.9. Would you expect the volatility of the 1-year forward price of oil to be greater than or less than the volatility of the spot price? Explain your answer.
- 33.10. What are the characteristics of an energy source where the price has a very high volatility and a very high rate of mean reversion? Give an example of such an energy source.
- 33.11. How can an energy producer use derivatives markets to hedge risks?
- 33.12. Explain how a  $5 \times 8$  option contract for May 2009 on electricity with daily exercise works. Explain how a  $5 \times 8$  option contract for May 2009 on electricity with monthly exercise works. Which is worth more?
- 33.13. Explain how CAT bonds work.
- 33.14. Consider two bonds that have the same coupon, time to maturity, and price. One is a B-rated corporate bond. The other is a CAT bond. An analysis based on historical data shows that the expected losses on the two bonds in each year of their life is the same. Which bond would you advise a portfolio manager to buy and why?

**Further Questions**

- 33.15. An insurance company's losses of a particular type are to a reasonable approximation normally distributed with a mean of \$150 million and a standard deviation of \$50 million. (Assume no difference between losses in a risk-neutral world and losses in the real world.) The 1-year risk-free rate is 5%. Estimate the cost of the following:
- (a) A contract that will pay in 1 year's time 60% of the insurance company's costs on a pro rata basis
  - (b) A contract that pays \$100 million in 1 year's time if losses exceed \$200 million.
- 33.16. How is the tree in Figure 33.2 modified if the 1- and 2-year futures prices are \$21 and \$22 instead of \$22 and \$23, respectively. How does this affect the value of the American option in Example 33.3.